

Attribution-quality metrics: MaRC and IG, baseline vs ABTT

Trimmed view for LaTa and PhilTa (main-text candidates).
 20 pairs per model from the phase12f curated visualization set.
 Generated 2026-04-20.

Method	Suff@25% $\uparrow$		Comp@25% $\uparrow$		Cmpct@0.8 $\downarrow$		ρ_{LOO} $\uparrow$	
	base	abtt	base	abtt	base	abtt	base	abtt
<i>LaTa</i>								
IG	0.677	0.202	0.158	0.038	0.302	0.383	-0.118	-0.007
MaRC	0.198	0.014	0.066	-0.061	0.855	0.935	-0.095	-0.080
<i>PhilTa</i>								
IG	0.763	0.562	0.226	0.091	0.144	0.252	-0.017	0.116
MaRC	-0.269	0.026	-0.011	0.009	0.676	0.940	0.023	0.058

Within each (method, metric) cell, the better of {baseline, ABTT} is **bolded**. Numbers are group means over 20 curated pairs per model.

Reading guide. **Suff@25%** ($\uparrow$): similarity retained when only the top 25% attributed tokens are kept. Higher is better. **Comp@25%** ($\uparrow$): similarity drop when those tokens are removed. Higher is better. **Cmpct@0.8** ($\downarrow$): fraction of tokens needed to recover 80% of the full-pair cosine. Lower is better. **ρ_{LOO}** ($\uparrow$): rank correlation between each attribution’s claimed importance and its true leave-one-out impact on the cosine. Higher is better.

Bottom line on this subset.

- ρ_{LOO} improves under ABTT on every (model, method) cell shown here, and on every cell in the wider 4-model $\times$ 9-method sweep (8/8). This is the one robust signal.
- The other three metrics (sufficiency, comprehensiveness, compactness) go the *other* way under ABTT for IG on both T5 models (3/4 metrics worse). RetrievalMarK on PhilTa improves 3/4. RetrievalMarK on LaTa improves 1/4.
- So the original sanity-check threshold “ABTT improves $\geq 3/4$ metrics for both MaRC and IG on at least one T5 model” is *not* met. Only PhilTa $\times$ MaRC clears it.
- Caveat: a fraction of the apparent “worsening” is that ABTT changes both the cosine being explained and the attribution scores themselves, so cross-variant deltas mix two effects. With only 20 pairs per cell the numbers are descriptive, not statistically powered.